Table 2: DeepReviewer-14B Evaluation of AI-Generated Papers from Various AI Scientist Systems. Scores reflect averages across the 'Num' of available papers. Note: Publicly available papers may be curated and not fully representative of typical system output.

AI Scientist System	Num	Soundness↑	Presentation↑	Contribution↑	Decision↑	Rating↑	Percentile↑
HKUSD AI Researcher	7	1.75	1.46	1.57	0.0	2.57	3.43%
AI Scientist	10	2.08	1.80	1.75	0.0	3.35	8.22%
AI Scientist v2	3	1.67	1.50	1.50	0.0	2.33	2.04%
CycleResearcher-12B	6	2.25	1.75	2.13	0.0	3.75	16.88%
Zochi	2	2.38	2.38	2.25	0.0	4.63	29.96%

3.3. LLM-as-a-Judge Reveals the Implementation Weaknesses

To further support the existence of implementation gap, we employ a simulated peer review methodology to assess the actual quality of scientific outputs from current AI Scientist systems, particularly their implementation-level reliability. We select 28 publicly available research papers generated independently by five different AI Scientist systems and utilize the SoTA review model DeepReviewer-14B (Zhu et al., 2025) to conduct systematic evaluation under unified standards. We acknowledge that potential selection bias in the public availability of these papers (e.g., researchers may only publish better-performing outputs) means our evaluation results may not fully represent the average output quality of these systems across all scenarios. Nevertheless, **this analysis provides valuable insights into the general quality level of current AI-generated research papers.**

Quantitative Assessment Results. As Table 2 demonstrates, DeepReviewer-14B assigns generally low average scores to these AI-generated papers across multiple core dimensions. The Rating scale ranges from 1-10, where a score of 6 indicates acceptable quality, while Soundness, Presentation, and Contribution scores range from 1-4. The highest-rated system, Zochi (with a sample size of 2 papers), achieves **an average rating of only 4.63**, while other systems score even lower, typically in the 2-3 point range. These quantitative scores reveal that current AI Scientist systems face **significant challenges in independently producing high-quality research papers.**

Table 3: Defect Categories and Their Issues.

Defect Category	Number	Percentage
Experimental Weakness	28	100%
Methodological Unclarity/Flaws	27	96.4%
Writing & Presentation Issues	26	92.9%
Novelty Concerns	25	89.3%
Theoretical Weakness	24	85.7%
Literature Review Deficiencies	22	78.6%
Practicality & Robustness Gaps	21	75.0%
Reproducibility Issues	20	71.4%
Computational Cost Concerns	18	64.3%
Component Analysis	16	57.1%
Hyperparameter Analysis Lacking	16	57.1%
Ethical Considerations Missing	3	10.7%

Table 3 shows that among the twelve major defect categories, "Experimental Weakness" appears across all 28 evaluated AI-generated papers, with a 100% occurrence rate. **This finding supports our positions regarding implementation capability limitations**, in experimental design, execution, and result analysis. The second and third most prevalent issues are "Methodological Unclarity/Flaws" (96.4%) and "Writing & Presentation Issues" (92.9%), which reflect AI Scientists' insufficient ability to clearly articulate and implement research plans. "Novelty Concerns" (89.3%) and "Theoretical Weakness" (85.7%) occur frequently, indicating that when AI Scientists generate complete papers, they struggle to propose original scientific contributions with solid theoretical foundations. The prevalence of these high-frequency defects highlights **systemic issues in the scientific rigor and implementation quality of current AI-generated research**, falling below the standards for reliable and valuable scientific outputs.

4. Rooted Limitations of Execution Capabilities

Our empirical analysis (Section 3) reveals a clear pattern that while AI Scientists are conceptualized as advanced iterations of traditional scientific tools, they consistently fail at implementation and verification procedures across diverse scientific contexts. This raises a critical question: Why do these sophisticated systems fail to achieve consistently strong results, especially when traditional scientific tools, wielded by human researchers, prove highly effective? To understand this paradox, we provide a discussion on the root cause of the implementation gap (Section 4.1) and present an in-depth analysis of the fundamental limitations of AI Scientist (Section 4.2).

4.1. Two Primary Facets of Implementation Gap

The implementation gap for AI Scientists comprises two primary facets: (1) **AI Scientists often exhibit bottlenecks in the planning and execution stages**. This manifests in three key areas: **failures in long-range logical reasoning** required for coherent experimental design, **inadequate multi-agent collaboration** capabilities including strategic planning across complex multi-file implementations and converting conceptual ideas into working code, and **insufficient coordination** with external tools and systems; (2) Even when implementation code is generated, **AI Scientists demonstrate fundamental weaknesses in evaluation processes.** This includes failures in debugging capabilities, experimental validation, result interpretation, and iterative refinement based on experimental feedback. Current systems lack robust mechanisms for assessing implementation quality, validating experimental outcomes, and providing reliable feedback loops that can guide subsequent implementation improvements.

Prevent building "castle in the air". Agent tools often produce difficult-to-verify code and experiments, while evaluation gaps prevent AI Scientists from recognizing and correcting implementation issues through iterative refinement. Without fundamentally enhancing both capabilities, the idealized AI Scientist capable of independent scientific exploration will remain inefficient.

4.2. Rooted Limitations

From the current literature on AI scientists, we conclude four major limitations that collectively explain why AI scientists struggle with complex, multi-stage implementation processes:

Limitation 1: fundamental cognitive and execution capabilities. Scientific implementation requires sophisticated long-range logical reasoning across multiple abstraction levels. Existing LLMs demonstrate significantly decreased coherence and robustness as reasoning chains extend (Wu et al., 2025b,a), and increased thinking time does not necessarily yield stronger performance (Ballon et al., 2025). Furthermore, LLM-based agents possess limited capacity to retain past interaction information, with memory deteriorating as text length increases (Pink et al., 2025, Cemri et al., 2025). Most critically, mainstream language models exhibit markedly weaker performance in multi-turn dialogues or multi-step interactive tasks requiring context coherence, deep understanding, and state tracking, with average performance decreases reaching 39% (Laban et al., 2025). This capability degradation in scenarios involving long-range dependencies and complex interactions directly **constrains AI Scientist performance in executing complex scientific experiments** requiring sustained attention and coherent reasoning chains.

Limitation 2: strategic planning and reasoning. Scientific implementation requires comprehensive

abilities for strategic reasoning, continuous monitoring, and dynamic adjustment across all research stages (Lu et al., 2024, Yamada et al., 2025). High-quality research implementation demands global planning abilities spanning entire codebases, which typically contain multiple interdependent files with hundreds of lines requiring coordinated modification (Jimenez et al., 2024, Aleithan et al., 2024). Long-term, complex scientific exploration tasks such as discovering new materials, and modeling complex biological systems particularly require continuous iteration of research directions and experimental strategies over extended time scales based on emerging results and external feedback (Merchant et al., 2023, Brixi et al., 2025, Weng et al., 2023). However, current LLMs demonstrate inadequate adaptive planning and metacognitive abilities when handling highly open, creative scientific research requiring dynamic adjustments to overall research blueprints. While reinforcement learning approaches may potentially enhance LLMs' generalization and metacognitive capabilities, the resource investment required for "inventor" roles like AI Scientists that need to perform complex asynchronous operations and real-world interactions proves enormous. Figure 4 highlights AI's acceleration over human performance in complex tasks such as reasoning and web-based research. While AI Scientists also achieve tasks faster than humans, their estimated single-sample RL training time is orders of magnitude greater than simpler AI agents. This substantial increase in required sampling time (detailed in Appendix A) underscores the immense challenge of developing AI Scientists via standard RL methodologies.

Limitation 3: multi-agent collaboration. Ideal AI Scientist should seamlessly integrate into complex research ecosystems, engaging in efficient, accurate interactions and coordination with human scientists, other AI agents, and external tools (Guo et al., 2024, Qian et al., 2024, Pu et al., 2025b). This requires AI Scientist to not only understand instructions conforming to collaborative protocols but also precisely execute the implementation phases assigned to it within tasks and reliably feed its outputs back to the collaborative network (Bo et al., 2024, Zhang et al., 2024). However, current LLM Agents still have considerable room for improvement in robustness and adaptability when interacting with dynamic environments (Wei et al., 2025). For instance, when calling a series of external APIs to complete a complex scientific computational process, LLM often struggles to handle subtle changes in API interfaces, and other practical engineering issues (Shen et al., 2025).

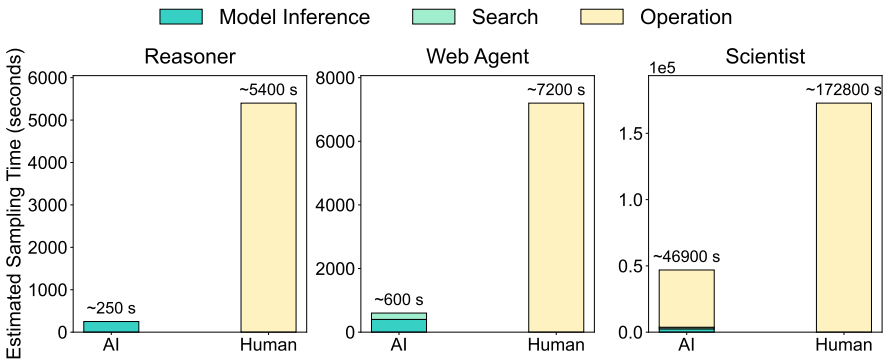


Figure 4: Estimated time to solve representative tasks for different agent types (AI vs. Human), which for AI agents also corresponds to single-sample RL sampling duration.

Limitation 4: evaluation and verification. Existing benchmarks such as MLE-Bench (Chan et al., 2024) and PaperBench (Starace et al., 2025) primarily focus on the complete reproduction of code and experiments from papers. SciReplicate-Bench (Xiang et al., 2025) emphasizes generating necessary code from scientific papers, while ScienceAgentBench (Chen et al., 2025) concentrates on independent and singular data-driven tasks. **However, there is currently a lack of a comprehensive benchmark that can evaluate the entire scientific workflow**, from initial idea generation through to final implementation and completion. This absence makes it difficult to fairly compare the end-to-end capabilities of different AI Scientist systems.